5-Whys Root Cause Analysis

Course Enrollment Failure

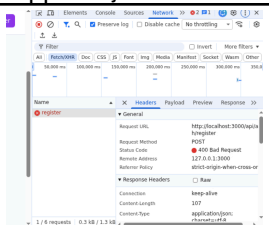
Instructions: Use the failed enrollment request captured in browser DevTools. Complete every field, make each Why follow logically from the previous response, and attach a readable Network-tab screenshot.

Tester	Gültəkin Əzizova	**Date**	2026-10-15
Test cycle	Course Enrollment — Sprint Execution	**Browser**	Microsoft Edge
Application	MindBridge	**URL**	http://localhost:3000

Observed symptom

When an invalid date format (2026-13-01) was entered for user U010 during course registration, the system failed to complete the registration and returned an unexpected error.

Technical evidence from DevTools

Request method and URL	POST http://localhost:3000/api/enroll
HTTP status code	400 Bad Request
Request payload	{"UserID": "U010", "FirstName": "Isabella", "LastName": "Taylor", "Email": "isabella.taylor@example.com", "CourseID": "C110", "EnrollmentDate": "2026-13-01"}
Response body	{"error": "Invalid date format or non-existent month supplied"}
Evidence screenshot	

5-Whys analysis

Level	Question	Evidence-based response
Why 1	Why did this happen?	Because the server did not accept the date 2026-13-01 and returned a 400 Bad Request error.
Why 2	Why did the previous condition occur?	This is because the 13th month does not exist in the calendar, and an error was detected during date validation on the backend.
Why 3	Why did the previous condition occur?	This is because the frontend form does not verify the validity of the month and day values entered by the user (client-side input validation).

Why 4	Why did the previous condition occur?	This is because date range limitations were not taken into account when coding the interface component.
Why 5	Why did the previous condition occur?	This is because date fields were not included in the list requiring strict validation checks during the code review process.

Root cause

Improper validation of the date format—specifically ensuring the month does not exceed 12—at both the frontend and API levels (lack of client/server input validation for calendar dates).

Corrective action

Adding date range validation to the frontend form and the backend API, as well as including this scenario in automated tests.

Verification plan

Retesting the Fix:

The QA engineer will re-enter the date format that previously caused an error (e.g., 2026-13-01 or invalid month/day values).

It will be verified that an error message appears immediately on the client-side (interface) before the form is submitted.

It will be confirmed that the API correctly rejects invalid dates when a similar request is sent at the server level (POST /api/enroll).

Regression Tests and Scope:

Date Formats and Boundary Values: Boundary values for the YYYY-MM-DD format (e.g., the 01–12 month range, day validity, and leap years) will be covered.

Related Scenarios: Previous successful test cases (re-enrollment, active/inactive statuses, and prerequisite checks—QEAR-T1 through QEAR-T4) will be re-executed for regression purposes to ensure the fix does not negatively impact other functionalities of the course enrollment module.